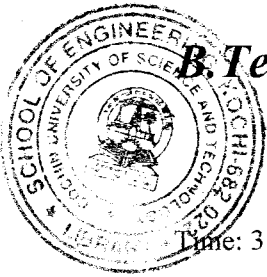


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B.Tech. Degree VII Semester Examination November 2018

ME 15-1701 REFRIGERATION AND AIR-CONDITIONING (2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Explain the term "Tonne of refrigeration".
- (b) "The COP of an air refrigeration cycle is very low, but still an air refrigeration system is most common in aircrafts": Discuss the statement.
- (c) Mention the advantages of vapour compression refrigeration system over air refrigeration system.
- (d) Discuss the function of absorber in vapour absorption refrigeration system.
- (e) "A completely odourless refrigerant is not desirable": Discuss the statement.
- (f) Discuss the relative advantages and disadvantages of air cooled and water cooled condensers.
- (g) Describe the working of an evaporative condenser.
- (h) Define the term 'bypass' factor used for cooling or heating coil and find an expression for the same.
- (i) Explain psychrometric chart with a neat sketch.
- (j) What do you understand by the term cooling load?

PART B

(4 × 10 = 40)

- II. (a) Draw the schematic of a boot-strap evaporative cycle of air refrigeration system, and show the cycle on T-s diagram. (7)
 - (b) Explain pulse tube refrigeration system with a neat sketch. (3)
- OR**
- III. (a) Describe the Bell-Coleman cycle and obtain the expression for the COP of the cycle. (5)
 - (b) A cold storage is to be maintained at -5°C while the surroundings are at 35°C. The heat leakage from the surroundings into the cold storage is estimated to be 29 kW. The actual C.O.P of the refrigeration plant is one third of an ideal plant working between the same temperatures. Find the power required to drive the plant. (5)

(P.T.O.)

- IV. (a) An ammonia ice plant operates between condenser temperature of 35°C and an evaporator temperature of -15°C . It produces 5 tonnes of ice per day from water at 25°C to ice at -5°C . The ammonia enters as dry saturated vapor and leaves the condenser as saturated liquid. Determine: (8)
- The capacity of the refrigerating plant
 - Mass flow of the refrigerant
 - Discharge temperature of ammonia from the compressor
 - Power of the compressor motor if the isentropic efficiency of the compressor is 85% and mechanical efficiency of the compressor is 90%
 - Relative efficiency
- The latent heat of formation of ice is 335 kJ/kg and specific heat of ice is 2.1 kJ/kg K .
- (b) Explain the advantages of vapour absorption system over vapour compression system. (2)
- OR**
- V. (a) Explain vapour absorption system and derive an expression for the COP of an ideal vapour absorption system in terms of temperature T_G at which heat is supplied to the generator, the temperature T_E at which heat is absorbed in the evaporator and the temperature T_C at which heat is discharged from the condenser and absorber. (5)
- (b) Explain Electrolux refrigerator system. State the advantages and disadvantages of Electrolux refrigerator over conventional refrigerators. (5)
- VI. (a) Write short notes on the types of refrigeration compressors. (5)
- (b) What is the necessity of intercooling in a multistage air compressor? Explain. (5)
- OR**
- VII. (a) Discuss the operation of a capillary tube in a refrigeration system. (4)
- (b) Explain in detail about the types of evaporators used in refrigeration system. (6)
- VIII. (a) Explain summer and winter air conditioning systems with neat sketches. (5)
- (b) Define the 'human comfort' and explain the factors which affect human comfort. (5)
- OR**
- IX. (a) List out the different types of loads to be considered for loading calculation of air conditioning equipment. (5)
- (b) An auditorium is to be maintained at a temperature of 26°C dry bulb and 50% relative humidity. Air is to be supplied at a temperature not lower than 15°C dry bulb. The sensible heat gain is 110 kW and the latent heat gain is 37.5 kW . Take ventilating air as 25% by weight of the air from the room and is at 35°C dry bulb and 60% relative humidity. Determine the refrigeration capacity in tons of refrigeration (TR). (5)